Meteorwave® 4000 - Dielectric Properties Table
AMMA Manufacturing Location

LAMINATE (Available from AGC U.S. manufacturing location – AMMA):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
0.0020	±	0.0005	1	1035				70%	0.0021	53	3.37	0.0018	3.34	0.0023	3.31	0.0026
0.0025	±	0.0005	1	1078				63%	0.0026	66	3.46	0.0019	3.40	0.0025	3.38	0.0028
0.0030	±	0.0005	1	1078				69%	0.0032	80	3.38	0.0018	3.35	0.0023	3.32	0.0026
0.0040	±	0.0005	2	1035				70%	0.0041	105	3.37	0.0018	3.34	0.0023	3.31	0.0026
0.0040	±	0.0005	1	2116				51%	0.0041	104	3.63	0.0022	3.53	0.0029	3.51	0.0032
0.0050	±	0.0007	1	2116				60%	0.0052	131	3.50	0.0020	3.43	0.0026	3.41	0.0029
0.0050	±	0.0007	2	1078				63%	0.0052	131	3.46	0.0019	3.40	0.0025	3.38	0.0028
0.0060	±	0.0007	2	1078				69%	0.0063	160	3.38	0.0018	3.35	0.0023	3.32	0.0026
0.0070	±	0.0010	2	1035	+	1	1078	67%	0.0067	171	3.40	0.0018	3.36	0.0023	3.34	0.0026
0.0080	±	0.0010	2	2116				51%	0.0082	208	3.63	0.0022	3.53	0.0029	3.51	0.0032
0.0100	±	0.0010	2	2116				60%	0.0104	263	3.50	0.0020	3.43	0.0026	3.41	0.0029
0.0140	±	0.0015	2	2116	+	1	2116	57%	0.0144	367	3.53	0.0020	3.46	0.0027	3.43	0.0030
0.0150	±	0.0015	3	2116				60%	0.0155	394	3.50	0.0020	3.43	0.0026	3.41	0.0029
0.0200	±	0.0025	5	2116				51%	0.0204	519	3.63	0.0022	3.53	0.0029	3.51	0.0032
0.0310	±	0.0040	6	2116				60%	0.0311	789	3.50	0.0020	3.43	0.0026	3.41	0.0029

Meteorwave® 4000 - Dielectric Properties Table
AMMA Manufacturing Location

PREPREG (Available from AGC U.S. manufacturing location - AMMA):

Glass	RC%	Thickness (inches)	Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
1035	72%	0.0022	57	3.34	0.0018	3.32	0.0022	3.30	0.0025
1035	77%	0.0028	70	3.29	0.0017	3.28	0.0020	3.26	0.0023
1035	79%	0.0030	77	3.27	0.0017	3.26	0.0020	3.24	0.0023
1078	74%	0.0038	97	3.32	0.0017	3.30	0.0021	3.28	0.0024
2116	57%	0.0048	121	3.54	0.0020	3.46	0.0027	3.44	0.0030

Meteorwave® 4000 - Dielectric Properties Table
AMMS Manufacturing Location

LAMINATE (Available from AGC Singapore manufacturing location – AMMS):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
0.0010	±	0.0005	1	1017				75%	0.0010	26	3.31	0.0017	3.29	0.0021	3.27	0.0010
0.0020	±	0.0005	1	106				75%	0.0021	54	3.31	0.0017	3.29	0.0021	3.27	0.0020
0.0020	±	0.0005	1	1035				70%	0.0021	53	3.37	0.0018	3.34	0.0023	3.31	0.0020
0.0025	±	0.0005	1	1078				63%	0.0026	66	3.46	0.0019	3.40	0.0025	3.38	0.0025
0.0030	±	0.0005	1	1078				69%	0.0032	80	3.38	0.0018	3.35	0.0023	3.32	0.0030
0.0035	±	0.0005	2	1035				64%	0.0034	86	3.44	0.0019	3.39	0.0025	3.37	0.0035
0.0040	±	0.0005	1	2116				51%	0.0041	104	3.63	0.0022	3.53	0.0029	3.51	0.0040
0.0040	±	0.0005	2	1035				70%	0.0041	105	3.37	0.0018	3.34	0.0023	3.31	0.0040
0.0050	±	0.0007	1	2116				60%	0.0052	131	3.50	0.0020	3.43	0.0026	3.41	0.0050
0.0050	±	0.0007	2	1078				63%	0.0052	131	3.46	0.0019	3.40	0.0025	3.38	0.0050
0.0050	±	0.0007	2	1035				75%	0.0050	128	3.31	0.0017	3.29	0.0021	3.27	0.0050
0.0060	±	0.0007	2	1078				69%	0.0063	160	3.38	0.0018	3.35	0.0023	3.32	0.0060
0.0070	±	0.0010	2	2013				56%	0.0072	183	3.55	0.0021	3.47	0.0027	3.45	0.0070
0.0080	±	0.0010	2	2116				51%	0.0082	208	3.63	0.0022	3.53	0.0029	3.51	0.0080
0.0080	±	0.0010	2	2013				61%	0.0083	211	3.48	0.0020	3.42	0.0026	3.40	0.0080
0.0100	±	0.0010	2	2116				60%	0.0104	263	3.50	0.0020	3.43	0.0026	3.41	0.0100

Meteorwave® 4000 - Dielectric Properties Table
AMMS Manufacturing Location

LAMINATE (Available from AGC Singapore manufacturing location – AMMS):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
0.0120	±	0.0015	2	2013	+	1	2116	58%	0.0124	315	3.53	0.0020	3.45	0.0027	3.43	0.0030
0.0120	±	0.0015	3	2013				61%	0.0124	316	3.48	0.0020	3.42	0.0026	3.40	0.0029
0.0150	±	0.0015	3	2116				60%	0.0155	394	3.50	0.0020	3.43	0.0026	3.41	0.0029
0.0280	±	0.0025	7	2116				51%	0.0286	727	3.63	0.0022	3.53	0.0029	3.51	0.0032
0.0300	±	0.0025	6	2116				60%	0.0311	789	3.50	0.0020	3.43	0.0026	3.41	0.0029
0.0590	±	0.0050	2	2116	+	10	2116	59%	0.0599	1522	3.51	0.0020	3.44	0.0026	3.42	0.0029

Meteorwave® 4000 - Dielectric Properties Table
AMMS Manufacturing Location

PREPREG (Available from AGC Singapore manufacturing location - AMMS):

Glass	RC%	Thickness (inches)	Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
106	75	0.0021	54	3.31	0.0017	3.29	0.0021	3.27	0.0024
106	77	0.0023	59	3.29	0.0017	3.28	0.0020	3.26	0.0023
106	80	0.0027	68	3.26	0.0016	3.26	0.0020	3.23	0.0023
1027	74	0.0016	39	3.32	0.0017	3.30	0.0021	3.28	0.0024
1035	70	0.0021	53	3.37	0.0018	3.34	0.0023	3.31	0.0026
1035	72	0.0022	57	3.34	0.0018	3.32	0.0022	3.30	0.0025
1035	75	0.0025	64	3.31	0.0017	3.29	0.0021	3.27	0.0024
1035	77	0.0028	70	3.29	0.0017	3.28	0.0020	3.26	0.0023
1035	79	0.0030	77	3.27	0.0017	3.26	0.0020	3.24	0.0023
1067	75	0.0027	68	3.31	0.0017	3.29	0.0021	3.27	0.0024
1067	77	0.0029	74	3.29	0.0017	3.28	0.0020	3.26	0.0023
1078	67	0.0029	75	3.40	0.0019	3.36	0.0023	3.34	0.0026
1078	69	0.0032	80	3.38	0.0018	3.35	0.0023	3.32	0.0026
1078	72	0.0035	90	3.34	0.0018	3.32	0.0022	3.30	0.0025
1078	74	0.0038	97	3.32	0.0017	3.30	0.0021	3.28	0.0024
1078	78	0.0046	116	3.28	0.0017	3.27	0.0020	3.25	0.0023

Meteorwave® 4000 - Dielectric Properties Table
AMMS Manufacturing Location

PREPREG (Available from AGC Singapore manufacturing location - AMMS):

Glass	RC%	Thickness (inches)	Thickness (µm)	2 GHz Dk	2 GHz Df	10 GHz Dk	10 GHz Df	40 GHz Dk	40 GHz Df
1080	67	0.0030	76	3.40	0.0019	3.36	0.0023	3.34	0.0026
1080	69	0.0032	81	3.38	0.0018	3.35	0.0023	3.32	0.0026
1080	72	0.0036	91	3.34	0.0018	3.32	0.0022	3.30	0.0025
1080	74	0.0039	99	3.32	0.0017	3.30	0.0021	3.28	0.0024
2013	60	0.0040	102	3.50	0.0020	3.43	0.0026	3.41	0.0029
2013	62	0.0043	109	3.47	0.0019	3.41	0.0025	3.39	0.0028
2116	57	0.0048	121	3.54	0.0020	3.46	0.0027	3.44	0.0030
2116	60	0.0052	131	3.50	0.0020	3.43	0.0026	3.41	0.0029

These laminate and prepreg items are representative of available materials. These items and their associated values may not be regularly available and additional lead times or minimum order quantities could apply. Please contact our office or your local technical representative for assistance with choosing the best material for your design and application.

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